Project Synopsis Template

1. Title: Used Car Prices Prediction

Team Details

Provide details of all four team members. The first student will be the Team Lead.

Student Name	College Name	Branch	Mobile No	Mail ID	Role (Lead/Member)
G.Akhila	Priyadarshini Institute of Science and Technology for women	CSE	9010035809	gundaakhila1303@gmail.com	Lead
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Program Track and Domain

Program Track (Engg-HUB / Engg-Spoke / Science-HUB / Science-Spoke / Commerce-Spoke)	Domain (Analytics / Machine Learning / Deep Learning / GenAI)
Engg-HUB	Machine Learning / Deep Learning / GenAI

Guide Details

Rapaka manoj Kumar[SME]

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2. Problem Statement

Used car prices vary depending on factors such as the car's brand, model, year, mileage, fuel type, condition, and location. We chose this problem because estimating the correct price of a used car can be difficult for both buyers and sellers, and prices may vary significantly in the market. This problem is important in real life because an accurate price estimate can help buyers make better purchasing decisions and help sellers set reasonable prices. The main gap is that traditional price estimation may depend on manual judgment and may not consider multiple factors together. Therefore, this project aims to develop a Machine Learning model that predicts the price of a used car based on its important features, providing a faster, more consistent, and data-driven price estimate.

3. Objectives

Here are clear and simple **Objectives** for your **Used Car Price Prediction using Machine Learning** project:

1. **To collect and preprocess used car data** such as brand, model, year, mileage, fuel type, and transmission.
2. **To identify the factors that affect used car prices** using data analysis and visualization.
3. **To train a machine learning model** to predict the selling price of used cars.
4. **To compare model performance** using evaluation metrics such as MAE, MSE, RMSE, and R^2 score.
5. **To provide accurate price estimates** that can help buyers and sellers make better pricing decisions.

4. Methodology / Implementation Plan

1. machine learning regression algorithms such as **Linear Regression, Decision Tree Regression, Random Forest Regression, and Gradient Boosting**.

2. **Model Evaluation:** Compare the models using **MAE, MSE, RMSE, and R^2 score** and select the model with the best performance.

3. **Prediction:** Use the selected model to predict the estimated selling price of a used car based on user-provided details.

4. **Tools and Technologies: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Jupyter Notebook**, and optionally **Flask/Streamlit** for deployment. **Data Collection:** Collect used car datasets containing features such as car brand, model, manufacturing year, mileage, fuel type, transmission, engine capacity, and selling price.

5. **Data Preprocessing:** Handle missing and duplicate values, remove irrelevant data, convert categorical features into numerical form, and prepare the dataset for machine learning.

6. **Exploratory Data Analysis:** Analyze the relationship between car features and prices using statistical analysis and visualizations.

7. **Feature Selection:** Identify the most important features that influence used car prices, such as vehicle age, mileage, brand, and fuel type.

8. **Model Development:** Train

5. Resources and Timeline

A. Hardware & Software Requirements

Examples:

- Laptop/PC (specification)
- OS
- Programming languages (Python, R, etc.)
- Tools: Power BI, Jupyter Notebook, VS Code, Azure, etc.

B. Dataset

For your **Used Car Price Prediction using Machine Learning** project, you can write the Dataset section like this:

- **Dataset availability:** Yes, the dataset is available.
- **Dataset source:** Kaggle – Used Car Price Prediction dataset.
- **Source link:** Kaggle Used Car Price Prediction Datasets
- **Dataset attributes:** Car brand, model, year, mileage, fuel type, transmission, engine capacity, ownership details, and selling price.

- **Data collection:** If additional data is required, used-car listings can be collected from publicly available online sources and combined with the existing dataset after removing duplicates and irrelevant records.
- **Purpose:** The dataset will be cleaned and used to train ML regression models for predicting the selling price of used cars.

C. Project Timeline

Phase	Task	Tentative Completion Date
Phase 1	Problem definition, literature survey, and requirement analysis for used car price prediction	**Week 1**
Phase 2	Dataset collection, data exploration, and feature selection for used car price prediction	**Week 2**
Phase 3	Data preprocessing, feature engineering, and ML model design & training	**Week 3**
Phase 4	Model evaluation, price prediction, and web application development	**Week 4**
Phase 5	Testing, result analysis, model performance comparison, documentation, and final submission	**Week 5**

6. Expected Outcomes / Conclusion

1. **A functional machine learning model** capable of predicting the selling price of used cars from vehicle features.
2. **Identification of key price factors** such as car age, mileage, brand, fuel type, transmission, and engine capacity.
3. **Comparison of multiple ML regression models** to select the model with the best prediction performance.
4. **Improved price prediction accuracy** through data preprocessing, feature selection, and model evaluation.
5. **A working prediction prototype** where users can enter car details and receive an estimated used car price.

7. References

Here is a **ready-to-use References section** for your **Used Car Price Prediction using Machine Learning** project:

References

1. N. Pal, P. Arora, D. Sundararaman, P. Kohli, and S. S. Palakurthy, **"How much is my car worth? A methodology for predicting used cars prices using Random Forest,"** arXiv, 2017. (arXiv)

2. R. Sanjay and Dr. P. N. Shiammala, **"Used Car Price Prediction Using Machine Learning,"** IJCOPE Journal, 2026. (IJCOPE)

3. M. Ahmed, G. Geetha Priyanaka, P. Mohana Priya, P. Mahadev, and M. Prasanna, **"Used Car Price Prediction Using Random Forest Regression,"** American Journal of AI Cyber Computing Management, 2026. (AJACCM)

4. R. Yuvaraj and Dr. K. Dharmarajan, **"Used Car Price Prediction System Using Machine Learning Regression Techniques,"** Zenodo, 2026. (Zenodo)

5. **Scikit-learn Documentation** – Regression algorithms, Random Forest Regressor, model evaluation, MAE, MSE, RMSE, and R^2 score.

6. **YouTube Tutorial:** *Prediction of Used Car Prices Using Artificial Neural Networks and Machine Learning*, Python machine-learning project. (YouTube)

7. **YouTube Tutorial:** *Used Car Price Prediction using Machine Learning with Source Code*, Projectworlds. (YouTube)